CS & IT ENGINEERING



Algorithms

Analysis of Algorithms



Lecture No.- 03

By- Aditya sir

Recap of Previous Lecture



Topic

Topic

Apriori Analysis

Step-count mtd

Order of magnitude

(approx)

→ [Best Case
Worst Case

Topics to be Covered



- Topic
- Topic
- Topic

Best Case, Worst Case

HW

Asymptotic Notations ☆☆



About Aditya Jain sir

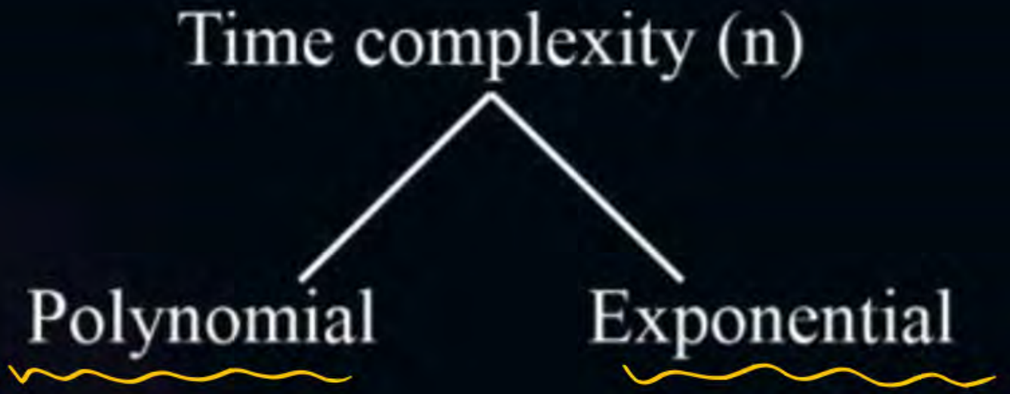
1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW



Topic : Asymptotic Notations



$$\text{☆}\ \boxed{B(n) \leq A(n) \leq W(n)}$$

$$\downarrow$$

$$B(n) = A(n) = W(n)$$

$B(n) \rightarrow$ Best Case

$A(n) \rightarrow$ Avg Case

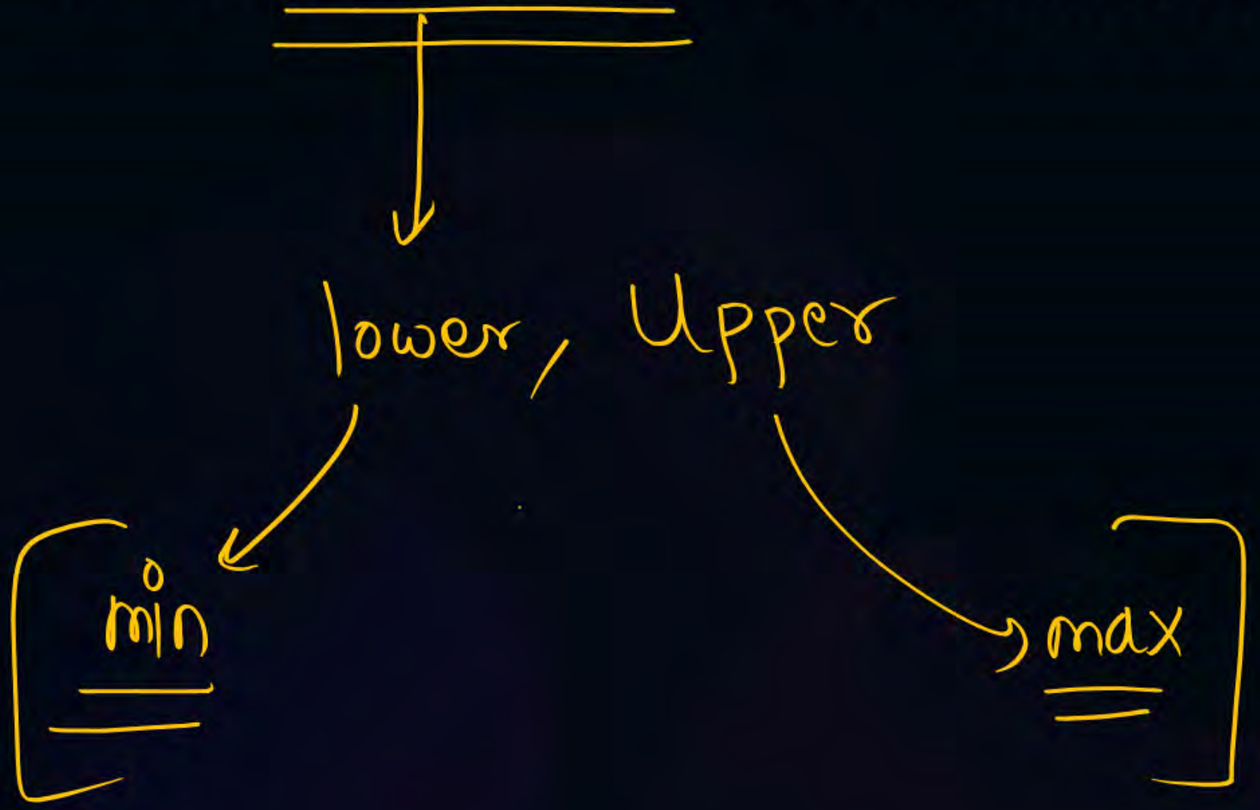
$W(n) \rightarrow$ Worst Case



Topic : Asymptotic Notations

Asymptotic Notations:

- The bounds/range of the function can be represented using these notations.





Topic : Asymptotic Notations

Types of Bounds:

(1) Upper Bound

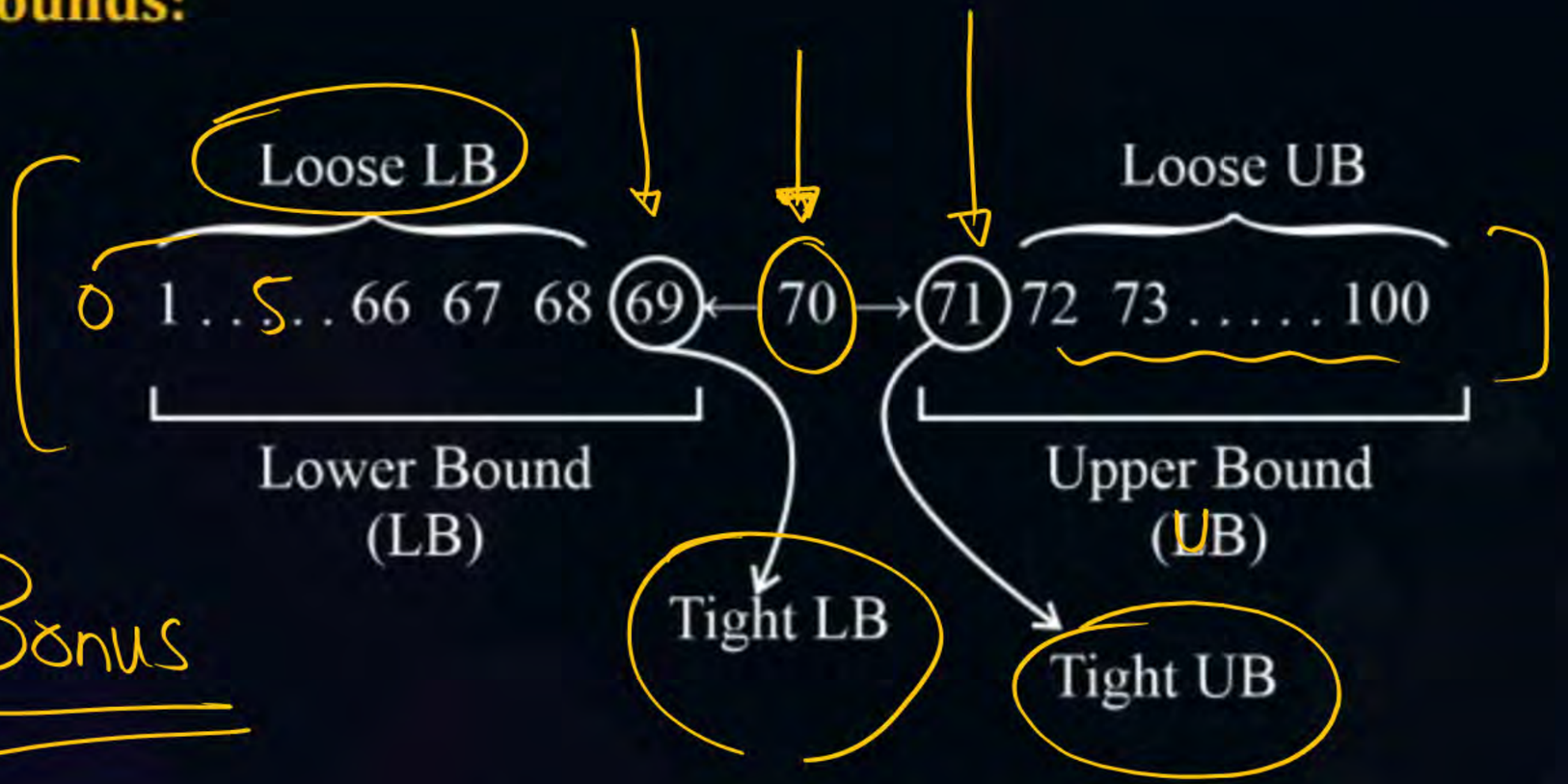
(2) Lower Bound

- (i) Lose Bound
- (ii) Tight Bound



Topic : Asymptotic Notations

Types of Bounds:



Bonus



Topic : Asymptotic Notations

V. V. Imp

Asymptotic Notations

(Tight + Lose Bound)

proper Lose Bound

(1) Big Notations

(i) UB : Big Oh → O

(ii) LB : Big Omega → Ω

(2) Small Notations

(iii) UB : Small oh → o

(iv) LB : Small omega → ω

(v) Theta : θ → Tight Bound

Oh → upper

Omega → lower

O, Ω, θ, o, ω



Topic : Asymptotic Notations

Whenever we determine the upper bound and lower bound, we should find that function 'g' which is closest to the given function.

Example:

Nagpur $\xrightarrow[\text{(Non-stop)}]{\text{Flight}}$ Delhi

LB $\left[\begin{array}{l} \text{P1} \rightarrow < 1 \text{ year} \\ \text{P2} \rightarrow < 1 \text{ week} \end{array}\right]$ UB

Tight UB $\left[\text{P3} \rightarrow < 5 \text{ hr}\right]$



Topic : Asymptotic Notations

Example2:

Tight Lower Bound



5 seater
AC
Automatic

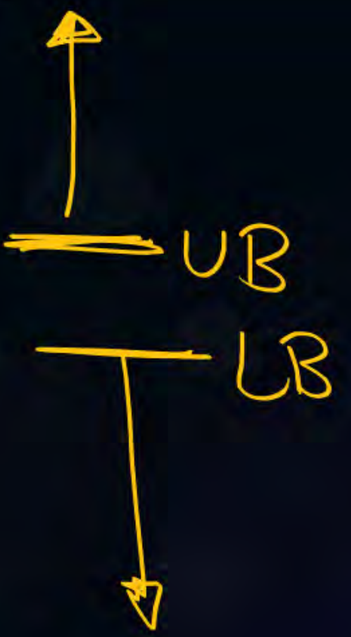
Friends

F1 : > 50 Rs
F2 : > 5000 Rs

Loose LB

F3 : > 5L Rs

Tight LB





Topic : Asymptotic Notations

Let 'f' and 'g' be functions from the set of integers/real to real number;

1. **Big-Oh(O):** Upper Bound (UB)

- f(n) is O(g(n)) if there exists some constant $c > 0$ and $n_0 > 0$ such that $f(n) \leq c.g(n)$, whenever $n \geq n_0$.

$f(n) = O(g(n))$ iff

$f(n) \leq c * g(n)$, Some $c > 0$

whenever $n \geq n_0$





Topic : Asymptotic Notations

Example:

(1) Order of Magnitude

$f(n) = n^2 + n + 1$

$1 + n \le n^2 + n^2$

$n = 2 \quad 1 + 2 \le 2^2 + 2^2$

$n = 3 \quad 1 + 3 \le 3^2 + 3^2$

$1 \le n^2$

$1 + n \le n^2 + n^2$

$1 + n + n^2 \le n^2 + n^2 + n^2$

$1 + n + n^2 \le 3n^2$

$f(n) \le c * g(n)$

Hence,

$f(n) = O(g(n))$

$1 + n + n^2 = O(n^2)$

$n \ge 1$



Topic : Asymptotic Notations

Shortcut:

Dominating term → Highest term with rate of growth

Example:

(i) $f(n) = n^2 + n + 1$

$f(n) = O(n^2)$

$n^2 + n + 1 \leq 3 * n^2$

(ii) $f(n) = 5n^3 + 8n + 7$

$f(n) = O(5n^3) = O(n^3)$

$O(5n^3) = O(n^3)$

Some

$5n^3 + 8n + 7 \leq 100n^3$



Topic : Asymptotic Notations

(1) Step-count method

$T(n) = 4n^2 + 8n + 8$

$T(n) = O(4n^2)$

$T(n) = O(n^2)$

(2) Order of magnitude

$T(n) = n^2 + n + 1$

$T(n) = O(n^2)$

$n^3 + n^2 + 1 = O(n^2)$ ⟶ False

$n^3 + n^2 + 1 = O(n^3)$ ⟶ True (Tight UB)

$= O(n^4)$

$= O(n^5)$

} ⟶ Lose UB



Topic : Asymptotic Notations

Example: n^2 vs 2^n

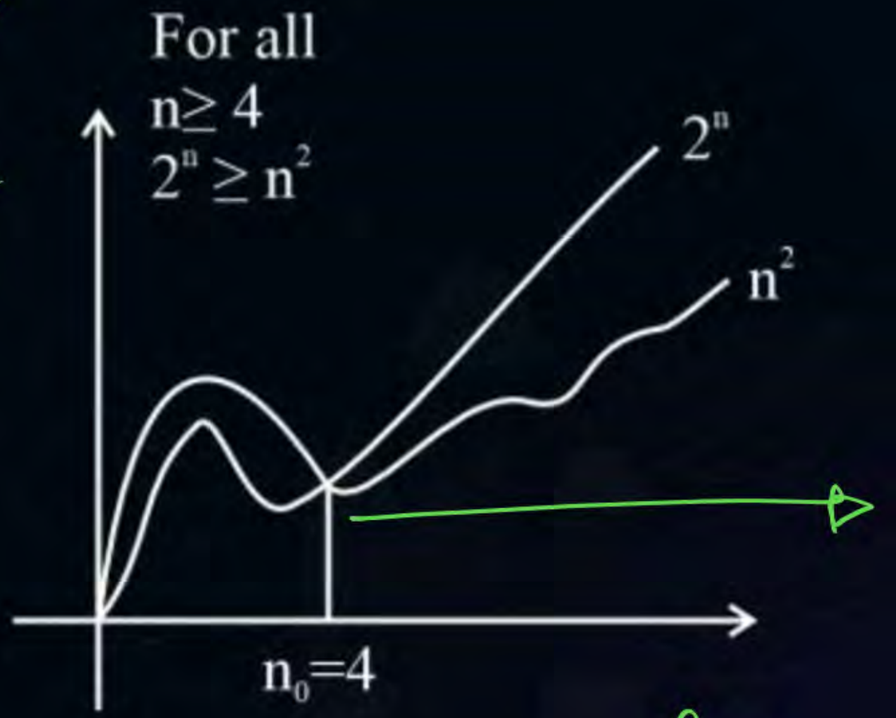
n	n^2	2^n
1	1	2
2	4	4
3	9	8
4	16	16
5	25	32
6	36	64
7	49	128

$n = 3$

$2^n > n^2$

$8 \geq 9$

$\times$



large value

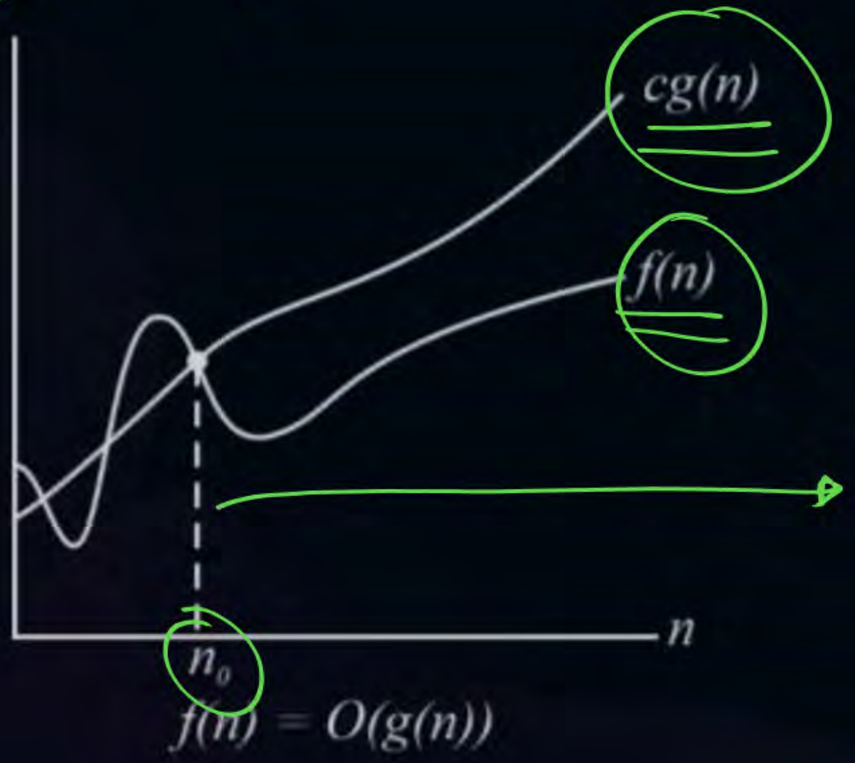


Topic : Asymptotic Notations

Big-Oh Notation:

$f(n) \leq c * g(n); n \geq n_0$

$f(n) = O(g(n))$



$f(n) \leq c * g(n)$



Topic : Asymptotic Notations

2. Big-Omega(Ω): Lower Bound (LB)

- f(n) in Ω(g(n)) iff there exists some constant 'c' and 'n_o' such that f(n) ≥ c.g(n), whenever n ≥ n_o.

c > 0

f(n) = Ω(g(n))

iff, f(n) ≥ c.g(n); n ≥ n_o

$f(n) \geq c * g(n)$ for some $c > 0$

then $f(n) = \Omega(g(n))$



Topic : Asymptotic Notations

Example: $8n^2 + 3n + 5$

$f(n) = 1 + n + n^2 \geq 1 * n^2 \rightarrow \Omega(n^2)$

$f(n) = 1 + n + n^2 \geq 1 * n \rightarrow \Omega(n)$

$f(n) = n + n + n^2 \geq 1 * \sqrt{n} \rightarrow \Omega(\sqrt{n})$

$f(n) = n + n + n^2 \geq 1 * 1 \rightarrow \Omega(1)$

$1 + n + n^2 \geq 1 * n^2$

$f(n) \geq c.g(n)$

$f(n) = \Omega(g(n)) = \Omega(n^2)$

Hence,

$1 + n + n^2 = \Omega(n^2)$

$8n^2 + 3n + 5 \geq 1 * n^2$

$f(n) \geq c * g(n)$

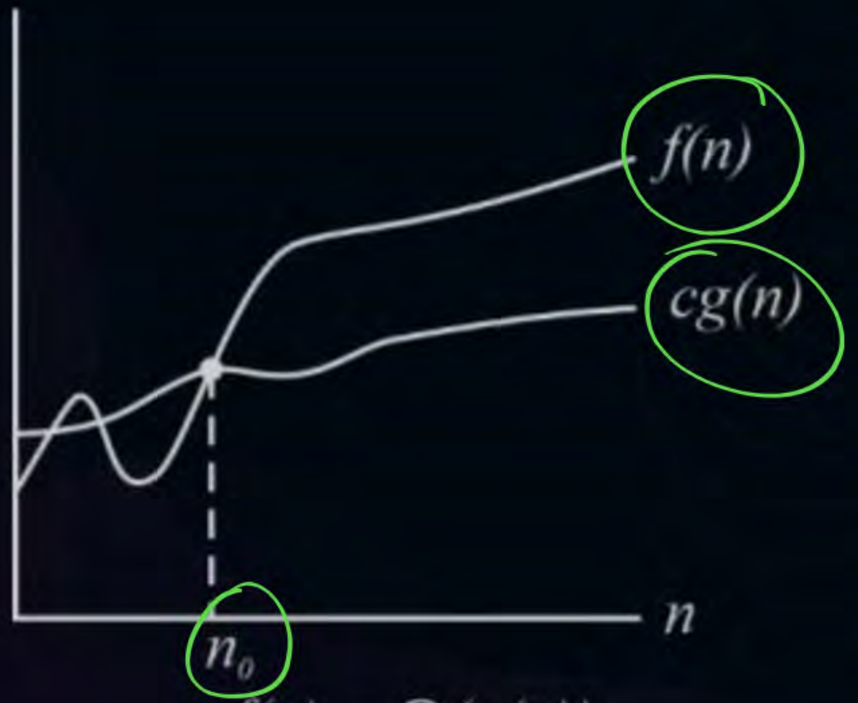


Topic : Asymptotic Notations

Big-Omega Notation:

$f(n) \geq c * g(n); n \geq n_0$

$f(n) = \Omega(g(n))$



$f(n) = \Omega(g(n))$

$f(n) \geq c * g(n)$



Topic : Asymptotic Notations

3. Theta(θ): Tight Bound

- f(n) is θ(g(n)) iff f(n) is O(g(n)) and f(n) is Ω(g(n))

$c_1 .g(n) \le f(n) \le c_2.g(n)$

If f(n) = O(g(n))
and
f(n) = Ω(g(n))
→ f(n) = θ(g(n))



Topic : Asymptotic Notations

Example: $n^2 + n + 1$

$1 * n^2 \leq 1 + n + n^2 \leq 3 * n^2$

$c_1 . g(n) \leq f(n) \leq c_2 . g(n)$

$f(n) = O(n^2)$

and

$f(n) = \Omega(n^2)$

$\Rightarrow f(n) = \theta(n^2) \rightarrow$ Tight bound

* TUB & TLB may or may not be same (later)

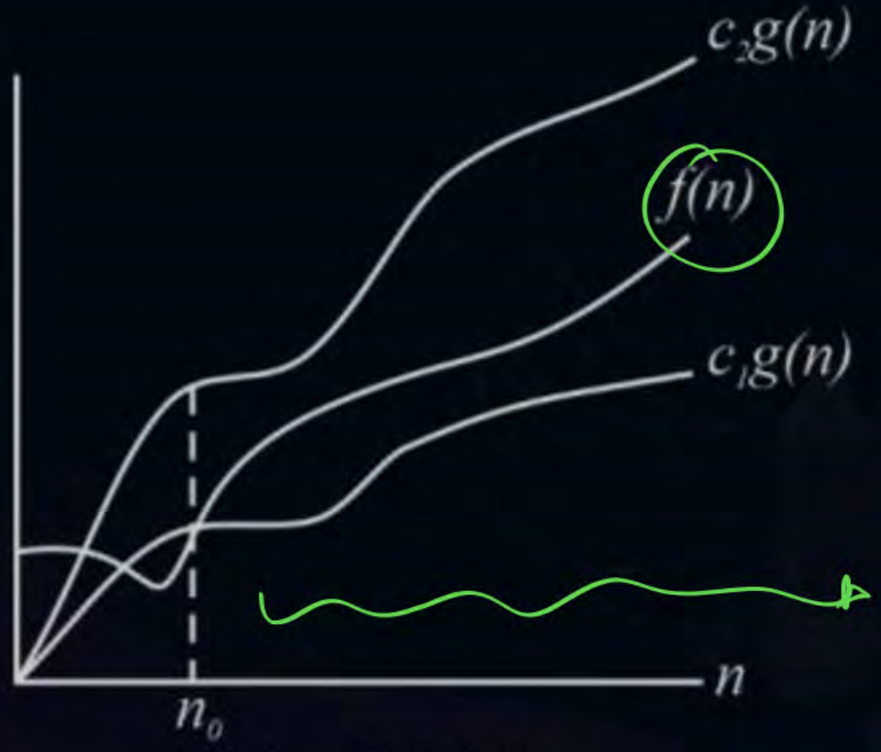


Topic : Asymptotic Notations

Theta Notation:

$c_1 * g(n) \leq f(n) \leq c_2 * g(n); n \geq n_0$

$f(n) = \theta(g(n))$

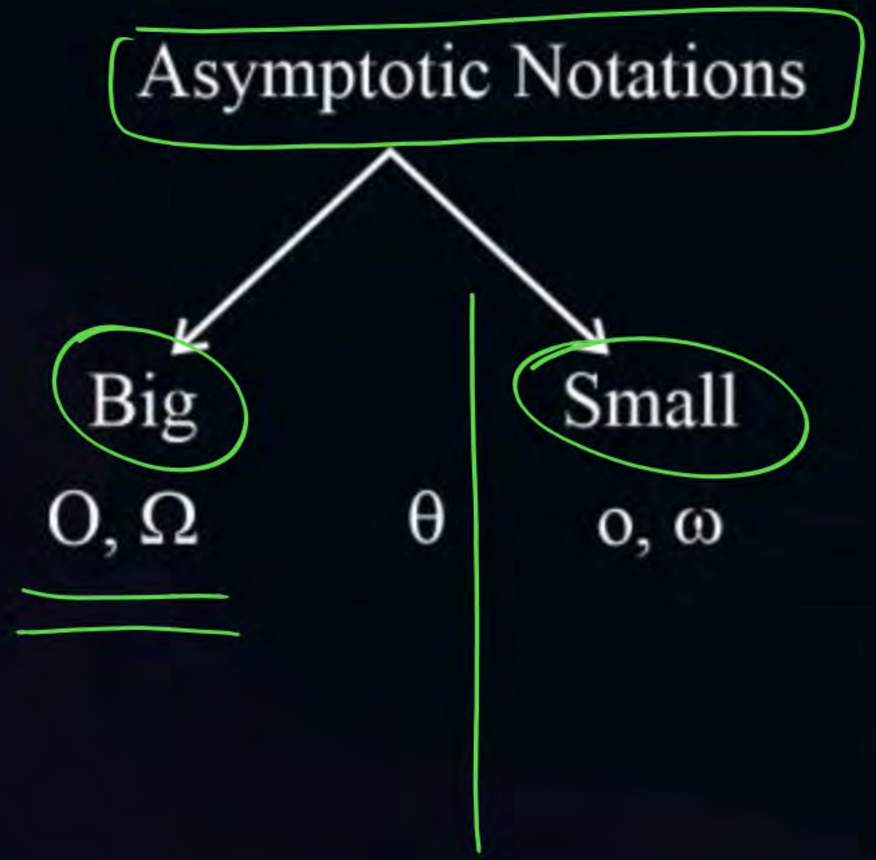


$f(n) = \Theta(g(n))$



Topic : Asymptotic Notations

Summary:



$n^2 = \Omega(n)$

$n^3 = \Omega(n^2)$

$2^n \geq c * n^2$

$2^n = \underline{\Omega}(n^2)$

$n = \underline{O}(n^2)$

$n^2 = O(n^3)$

Theta (Θ): tight Bound

$f(n) = 5n^2$ → $\Omega(n^2)$, $O(n^2)$ → $\Theta(n^2)$

$g(n) = 8n^2$ → $\Omega(n^2)$, $O(n^2)$ → $\Theta(n^2)$

$f(n) \underset{A}{=} g(n)$

equal rate of growth

Asymptotic

Mathematically

$5n^2 \stackrel{?}{=} 8n^2$

✗

Practice:

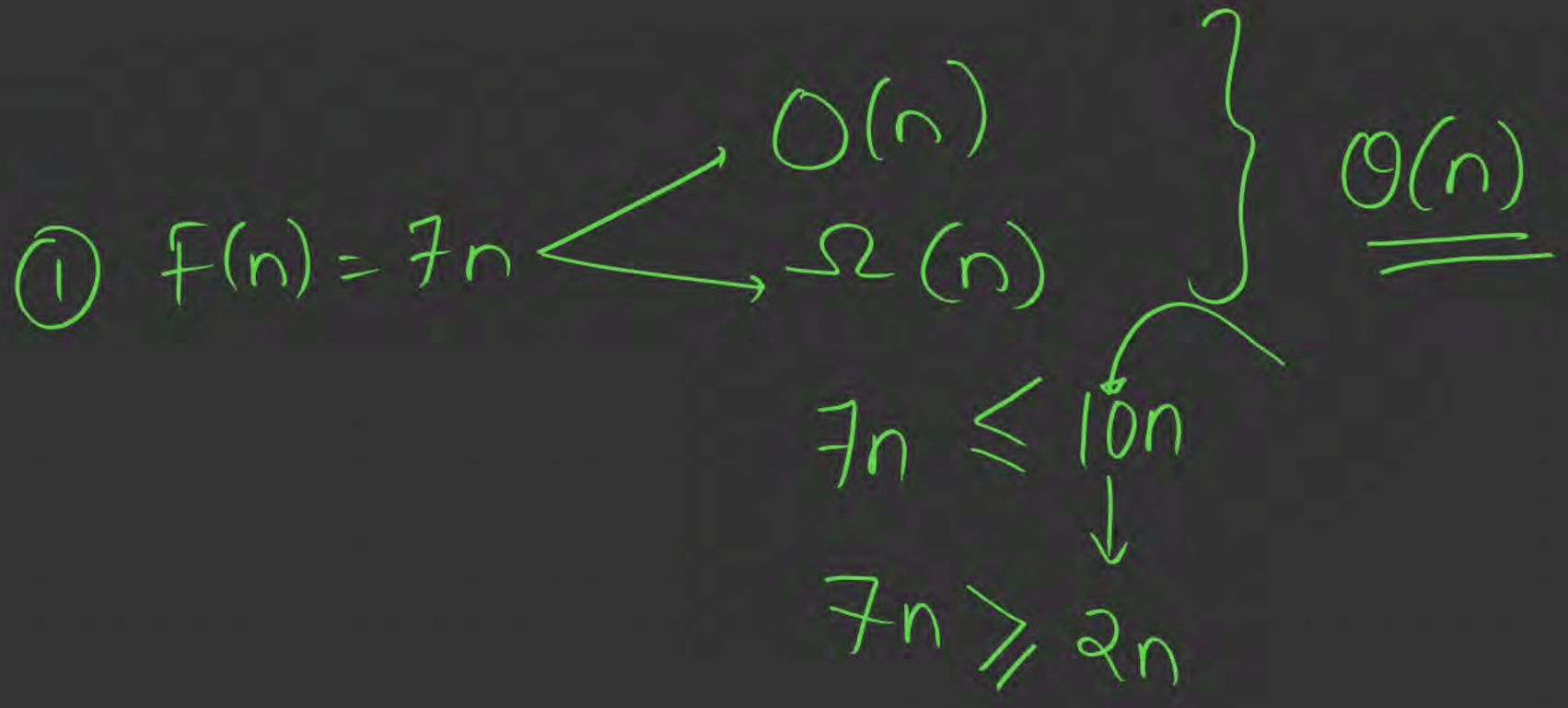
1. $f(n) = 7n$
2. $f(n) = 9^{150}$
3. $f(n) = 500 * \sqrt{n} + 2n^2 + 7$
4. $f(n) = 300 \log_2 n + 5\sqrt{n}$
5. $f(n) = 100 + 20n + 3n^2$

$\Omega(n^2)$
$O(n^2)$
$\Theta(n^2)$

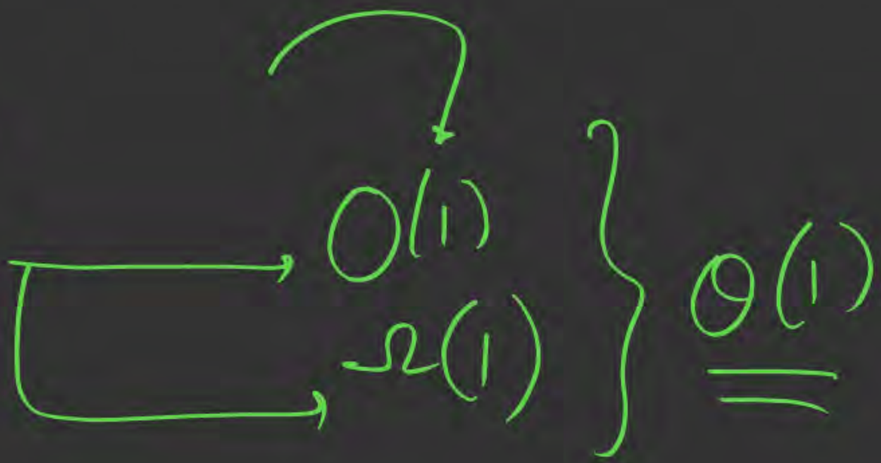
dominating term

$\Theta(\sqrt{n})$

$\Theta(n^2)$


① f(n) = 7n
O(n)
Ω(n)
Θ(n)
7n ≤ 10n
7n ≥ 2n

② $f(n) = 9^{150}$



$\Theta(1)$

independent of n

$\log n < \sqrt{n}$

$\log_2 n$ vs $\sqrt{n}$ → Tomorrow

Shortcut

$n = 64$

$\log_2(64)$ $\sqrt{64}$

$6 < 8$



Summary

→ B, A, W

→ O, Ω, O

Examples



THANK - YOU